Nonpolynomial coefficients

Sometimes we'll want to find a series solution to a differential equation with nonpolynomial coefficients.

In this case, we'll use a slightly different method to find the series solution, which will have us using power series multiplication.

Let's do an example so that we can see what the process looks like.

Example

Find the first three terms of the power series solutions of the differential equation around the ordinary point $x_0 = 0$.

$$y'' + (\sin x)y = 0$$

We know $x_0 = 0$ is an ordinary point, because $\sin x$ is analytic there. And we know that the Maclaurin series representation of $\sin x$ is

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots$$

So we can substitute into the differential equation to get

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots \right) \sum_{n=0}^{\infty} c_n x^n = 0$$

Now we'll just expand each series through its first few terms,

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$$(2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + \dots)$$

$$+\left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots\right)(c_0 + c_1x + c_2x^2 + c_3x^3 + \dots) = 0$$

$$(2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + \dots)$$

$$+x(c_0 + c_1x + c_2x^2 + c_3x^3 + \dots)$$

$$-\frac{1}{3!}x^3(c_0 + c_1x + c_2x^2 + c_3x^3 + \dots)$$

$$+\frac{1}{5!}x^5(c_0 + c_1x + c_2x^2 + c_3x^3 + \dots)$$

$$-\frac{1}{7!}x^7(c_0 + c_1x + c_2x^2 + c_3x^3 + \dots) + \dots = 0$$

$$(2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + \dots)$$

$$+(c_0x + c_1x^2 + c_2x^3 + c_3x^4 + \dots)$$

$$-\left(\frac{1}{3!}c_0x^3 + \frac{1}{3!}c_1x^4 + \frac{1}{3!}c_2x^5 + \frac{1}{3!}c_3x^6 + \dots\right)$$

$$+\left(\frac{1}{5!}c_0x^5 + \frac{1}{5!}c_1x^6 + \frac{1}{5!}c_2x^7 + \frac{1}{5!}c_3x^8 + \dots\right)$$

$$-\left(\frac{1}{7!}c_0x^7 + \frac{1}{7!}c_1x^8 + \frac{1}{7!}c_2x^9 + \frac{1}{7!}c_3x^{10} + \dots\right) + \dots = 0$$

and then collect equivalent powers of x.

$$2c_2 + (6c_3 + c_0)x + (12c_4 + c_1)x^2$$

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$$+\left(20c_5+c_2-\frac{1}{3!}c_0\right)x^3+\left(30c_6+c_3-\frac{1}{3!}c_1\right)x^4$$

$$+\left(42c_7+c_4-\frac{1}{3!}c_2+\frac{1}{5!}c_0\right)x^5+\left(56c_8+c_5-\frac{1}{3!}c_3+\frac{1}{5!}c_1\right)x^6+\ldots=0$$

From this result, we get a system of equations.

$$2c_2=0$$

$$6c_3+c_0=0$$

$$12c_4+c_1=0$$

$$20c_5+c_2-\frac{1}{3!}c_0=0$$

$$30c_6+c_3-\frac{1}{3!}c_1=0$$

$$42c_7+c_4-\frac{1}{3!}c_2+\frac{1}{5!}c_0=0$$

$$56c_8+c_5-\frac{1}{3!}c_3+\frac{1}{5!}c_1=0$$

which simplifies to

$$c_2=0$$

$$c_3=-\frac{1}{6}c_0$$

$$c_4=-\frac{1}{12}c_1$$

$$c_5=\frac{1}{120}c_0-\frac{1}{20}c_2$$

$$c_6=\frac{1}{180}c_1-\frac{1}{30}c_3$$

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$$c_7 = -\frac{1}{5{,}040}c_0 + \frac{1}{252}c_2 - c_4 \qquad c_8 = -\frac{1}{6{,}720}c_1 + \frac{1}{336}c_3 - c_5$$

Using earlier terms to simplify later terms, we get

$$c_2 = 0$$

$$c_3 = -\frac{1}{6}c_0 \qquad c_4 = -\frac{1}{12}c_1$$

$$c_5 = \frac{1}{120}c_0 \qquad c_6 = \frac{1}{180}c_0 + \frac{1}{180}c_1$$

$$c_7 = -\frac{1}{5{,}040}c_0 + \frac{1}{12}c_1 \qquad c_8 = -\frac{89}{10{,}080}c_0 - \frac{1}{6{,}720}c_1$$

Pull these coefficients together into the power series solution.

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + c_7x^7 + \dots$$

$$y = c_0 + c_1x - \frac{1}{6}c_0x^3 - \frac{1}{12}c_1x^4 + \frac{1}{120}c_0x^5 + \left(\frac{1}{180}c_0 + \frac{1}{180}c_1\right)x^6$$

$$-\left(\frac{1}{5{,}040}c_0 - \frac{1}{12}c_1\right)x^7 - \left(\frac{89}{10{,}080}c_0 + \frac{1}{6{,}720}c_1\right)x^8 + \dots$$

$$y = c_0 - \frac{1}{6}c_0x^3 + \frac{1}{120}c_0x^5 + \frac{1}{180}c_0x^6 - \frac{1}{5{,}040}c_0x^7 - \frac{89}{10{,}080}c_0x^8$$

$$+c_1x - \frac{1}{12}c_1x^4 + \frac{1}{180}c_1x^6 + \frac{1}{12}c_1x^7 - \frac{1}{6{,}720}c_1x^8 + \dots$$

$$y = c_0\left(1 - \frac{1}{6}x^3 + \frac{1}{120}x^5 + \frac{1}{180}x^6 - \frac{1}{5{,}040}x^7 - \frac{89}{10{,}080}x^8 + \dots\right)$$

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$$+c_1\left(x-\frac{1}{12}x^4+\frac{1}{180}x^6+\frac{1}{12}x^7-\frac{1}{6{,}720}x^8+\ldots\right)$$

Simplifying to just the first three terms in each series, we get the general solution to the differential equation.

$$y(x)=c_0\left(1-\frac{1}{6}x^3+\frac{1}{120}x^5+\ldots\right)+c_1\left(x-\frac{1}{12}x^4+\frac{1}{180}x^6+\ldots\right)$$

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